Ogallala aquifer groundwater-buffering project: what has been done in this folder

Folder **G:\MSU_GWB\datasets** (git repository, 9 commits) · state on 21 Oct 2026 · written after reading the README, the dataset catalogue, the TBI and residual-ET documents, both data-audit reports, 22 Colab notebooks, 18 local scripts, the *residual_et* package and the folder listings. Numbers are quoted from those files; items I could not confirm are marked.

1. Purpose and holdings

Satellite, reanalysis, model-assimilated and in-situ data over the aquifer (boundary *HPA_polygon/hp_bound2010*) were assembled to quantify how groundwater buffers vegetation and evapotranspiration (ET) in drought through a Temporal Buffering Index, **B(t) = F(t) − F_{exp}(t)**, where F is the ET anomaly and F_{exp} the anomaly expected from rain, root-zone soil moisture, VPD, radiation, temperature, land cover and season; a static vulnerability index VI = DSI / (α·SBI + β·GBI) sits beside it. Products live in *merged_datasets/* (31 entries, ~41 GB), *zarr/* (12 GB), *residual_et_store/* (2.8 GB). Files are standalone CF NetCDF/GeoTIFF at native resolution; sources are read-only and every derived product is a new file.

2. Data processing (22 Colab notebooks + 18 local scripts)

Weather, drought	gridMET daily 1979–2026-10-05 and drought indices (SPI/EDDI/…) from the lab's own files on the native 287×233 (1/24°) grid, packed int16; replaced the Earth Engine copies after the audit. Zarr copies of gridMET, drought, SMAP, SSEBop, MCD12Q1.
ET, energy	MODIS/SSEBop dekadal ET; ECOSTRESS ET (ECO_L3T_JET, 70 m tiles); ECHO-ET (UNH); **OpenET v2.1 monthly 30 m** (ensemble + 6 models, version switch to v2.0).
Carbon, SIF	OCO-2 and OCO-3 L2 Lite SIF soundings; GOSIF 8-day/monthly with its extraction mesh.
Water, soil moisture	SMAP L4 soil moisture (9 km) and L3 VOD; GRACE/GRACE-FO storage; HRES water-table depth; HiHydroSoil groups; gSSURGO stack (120 m) and plant-available water.
Vegetation, land	MOD13A3 vegetation indices, MCD12Q1 land cover, GLO DEM, GDE map, GFSAD1000 cropland, global irrigation areas, **USDA CDL** (2008→, on the 30 m Landsat tiles), **LANID** irrigation (30 m annual 1997–2017 + derived layers), **GEDI L4B** biomass (1 km).
Wells, towers	USGS well networks → water-level-change grids (2017–19, predevelopment–19), hydraulic conductivity, wells time series; AmeriFlux/NEON tower records.
How	Google Colab + Earth Engine exports for most; local Python (SSM env) for USGS, GOSIF, gSSURGO, clipped GDE and drought-fit products. Every notebook checks downloads, reads files back, resumes after a stop, and records its settings in the output.

3. Landsat indices (four generations of notebooks)

- **500 m per-scene** (*Landsat_Indices_Ogallala*): raw pixels pulled to Colab, indices computed locally, because Earth Engine compute cost was too high; resumes from per-group checkpoints. Only 1 of 59 path/rows is in *merged_datasets* (audit).
- **4 km via USGS ESPA** (*Landsat_ESPA_Indices_4km*), tested against a stand-in only.
- **30 m monthly** (*Landsat_Indices_30m_Monthly*): Landsat 5/7/8/9 Collection 2 Level-2 Tier 1, 18 indices (NDVI, EVI, EVI2, SAVI, MSAVI, OSAVI, GNDVI, NIRv, kNDVI, NDMI, NMDI, MSI, GVMI, NDWI, MNDWI, AWEInsh, AWEIsh, WI2015). Per pixel the **monthly median** of all passes of a UTM zone, no resampling, 5120-px tiles, yearly files *<year>/<tile>/*, int16 with scale factors, zlib 2. Choosable start/stop month; cloud, saturation, valid-range and aerosol screens; Landsat 7 cut at 2016; TM/ETM+→OLI harmonisation switch (on); extra variables *n_obs*, *obs_day*, *n_obs_oli*, NDVI/NDMI range. Verified by independent recomputation and on real Earth Engine windows; a full-record run is not documented here.
- **Maps** (*Landsat_Indices_30m_Monthly_Maps*) average a month to 4 km; six July-2020 figures are in *landsat indices plot/*.
- OpenET and CDL are placed on the same 30 m tiles as these files (pixel for pixel); LANID keeps its own 30 m Albers grid, which is the CDL's.

4. ET residual models (residual_et package, document of 7 Oct)

Two models estimate the expected ET anomaly whose residual is B: a **cross-fitted XGBoost** (25 inputs, 10 land-cover fractions, conformalised 5/50/95 % quantiles) and a **differentiable HBV water balance whose 15 parameters come from a 2-layer LSTM** (815,887 weights). Neither sees water-table depth or irrigation. Evaluation uses **5-fold rotation over half-year blocks** (70/20/10 train/test/validation, dry/normal/wet balanced) with an autocorrelation-based embargo.

- **Audit:** 29 findings in all seven requested bug classes, among them a drought file with no 2020 (future values leaked into 2020), in-sample residuals and placebo years, a spatial instead of per-pixel anomaly, exposure variables as inputs, and an LSTM that restarted at epoch 1 on every resume. The earlier headline numbers (XGBoost R^2 0.654, τ = +1.62, “null” placebo) **should not be used**.
- **Corrected package:** 24 Python files (sources, grid, climate, drought definitions, splits, uncertainty, both trainers, buffering statistics, tower comparison), **41 tests pass** (as of 7 Oct); a 144-cell smoke tile runs prepare → XGBoost → LSTM (train, resume, infer) → buffering.
- **Not done** (as of 7 Oct): full-aquifer training, by agreement. Open for you: the management-class thresholds and the dry-down rule.
- **Tower check** (run on real data): SSEBop vs 21 flux towers, 2,797 site-dekads, RMSE 12.0 mm/dekad, r = 0.74; May–Sep RMSE 15.9.

5. Quality control

- **Data audits** (6 and 20 Oct, *data audit/*); the 20 Oct one covered 49 files, 53.7 GB, 34 NetCDF, read lazily: **no file is physically corrupt**. Defects found: a 370 km² zero-elevation hole in the DEM, a provider-side repeat of three years in *eddi2y*, an undeclared -4.6×10^{12} value in the OCO files, unfinished exports (OCO-2, Landsat, ECOSTRESS, two gridMET copies, the last two no longer in the folder) and derived 4 km grids not aligned with gridMET.
- **Fix round (20 Oct):** SMAP, GFSAD, irrigation, HSG and MOD13A3 notebooks and the USGS 4 km scripts rebuilt → *…_reprocessed_20261020* files with date and version metadata; found on the way that the shared export step skipped 22 % of columns (fixed) and that Earth Engine's MODIS grid is a sphere, not WGS 84. MCD12Q1, HRES-WTD, SSEBop and GRACE share the grid flaw and were not touched.

6. Analysis, documents and plots

- *TBI_DOCUMENTATION.md* (62 kB): VVIP-style vulnerability index and its time-varying extension, with v1–v3 reruns on the USGS data (its XGBoost/LSTM results are superseded). *temporal buffering/*, *XGB model/* and *gw_buffering_xgb/* hold the earlier scripts and outputs.
- *Groundwater_Buffering_Research_Plan_v2_Ogallala.pdf* (20 Oct, 25 pages): three papers; dynamic, vegetation- and ET-driven irrigation masks; a rain-only reference model; 480 m blocks (16×16 Landsat pixels). 26 reference files in *gwb papers/*.
- *plots/* (34 files: shallow/deep groundwater clusters, ECOSTRESS tiles, GDE, MODIS/SSEBop, USGS products) and 7 plotting codes.

7. Open items

- Full runs of the 30 m Landsat, OpenET, CDL, LANID and GEDI notebooks: each was tested on small windows or stand-ins and I cannot see Drive, so completion is not confirmed (a 0.6 GB LANID file is already in *merged_datasets*).
- Residual-ET full training, then the buffering statistics; unfinished exports and misaligned 4 km grids from the audit.
- Newest notebooks (OpenET, CDL, LANID, GEDI) and this month's audit fixes are mostly not yet in the README or catalogue.

Layout: *Colab notebooks/* (22) · *local data processing/* (18) · *residual_et/* + *residual_et_store/* · *merged_datasets/* · *zarr/* · *data audit/* · *plots/*, *plotting codes/* · *gwb papers/* · *HPA_polygon/*. Written 21 Oct 2026 by Claude Code from the files above; not a replacement for them.